MODULE 10, LESSON 2

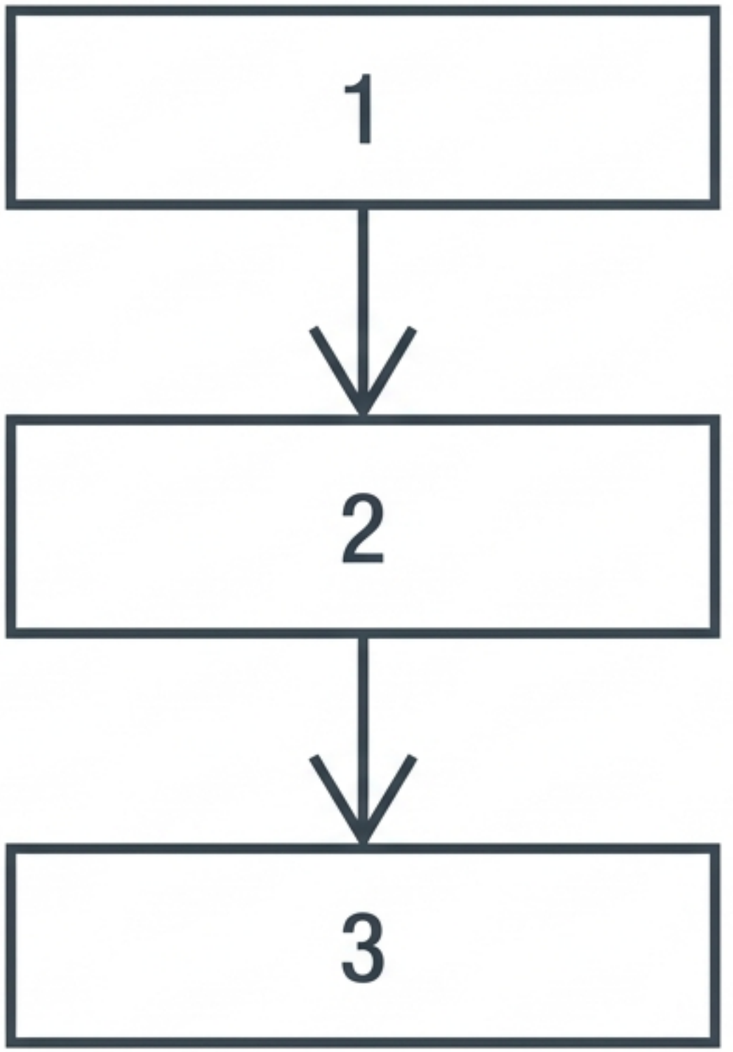
AI 101 FOR MECHANICAL ENGINEERS

UNDERFITTING, OVERFITTING AND REGULARIZATION

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COURSE OUTLINE

1. MODEL FITTING ERRORS
2. ALTERNATIVE ERROR CAUSES
3. REMEDIATION STRATEGIES



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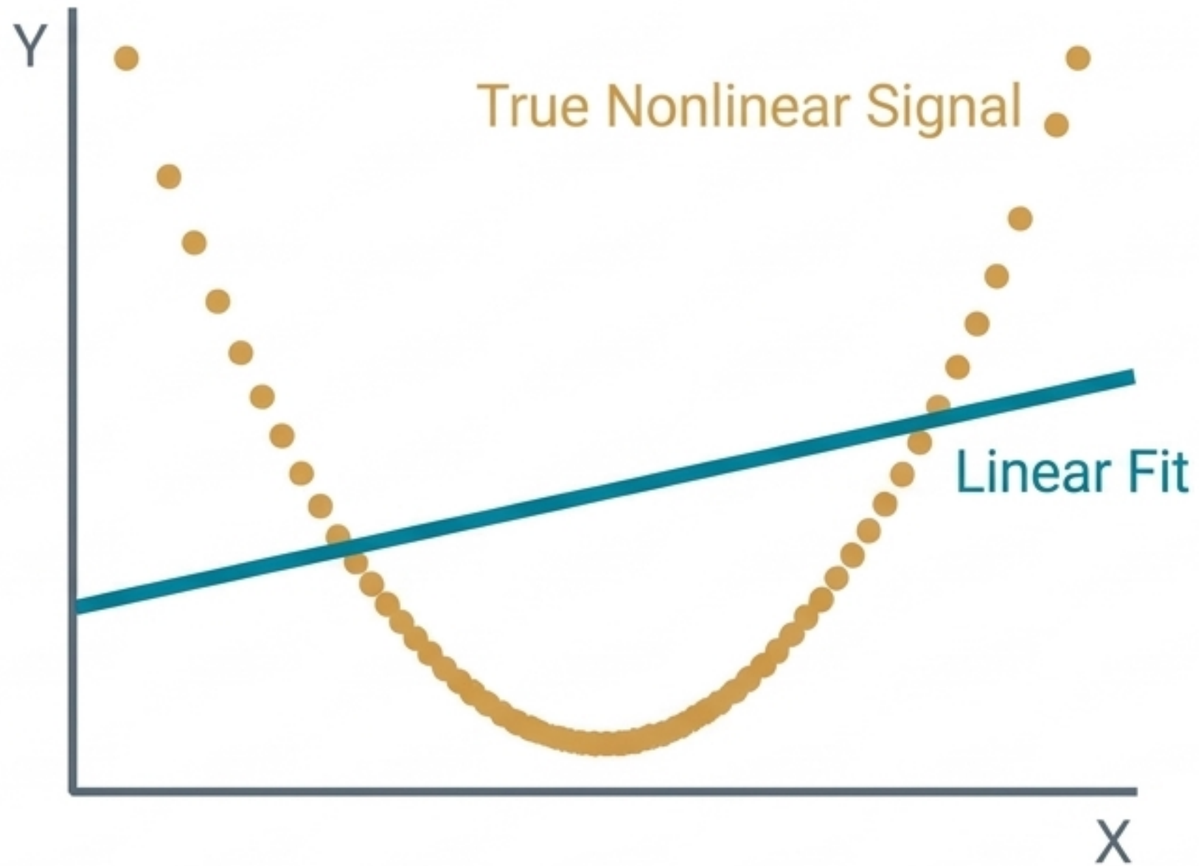
1. Model Fitting Errors

Define primary failure states of algorithmic learning representations.

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Underfitting Fails to Represent True Relationships

- Indicates insufficient representation of relevant data relationships.
- Model fails to capture strong nonlinear target responses.
- Produces high errors relative to established baselines.
- Demonstrates weak feature extraction or insufficient training capacity.



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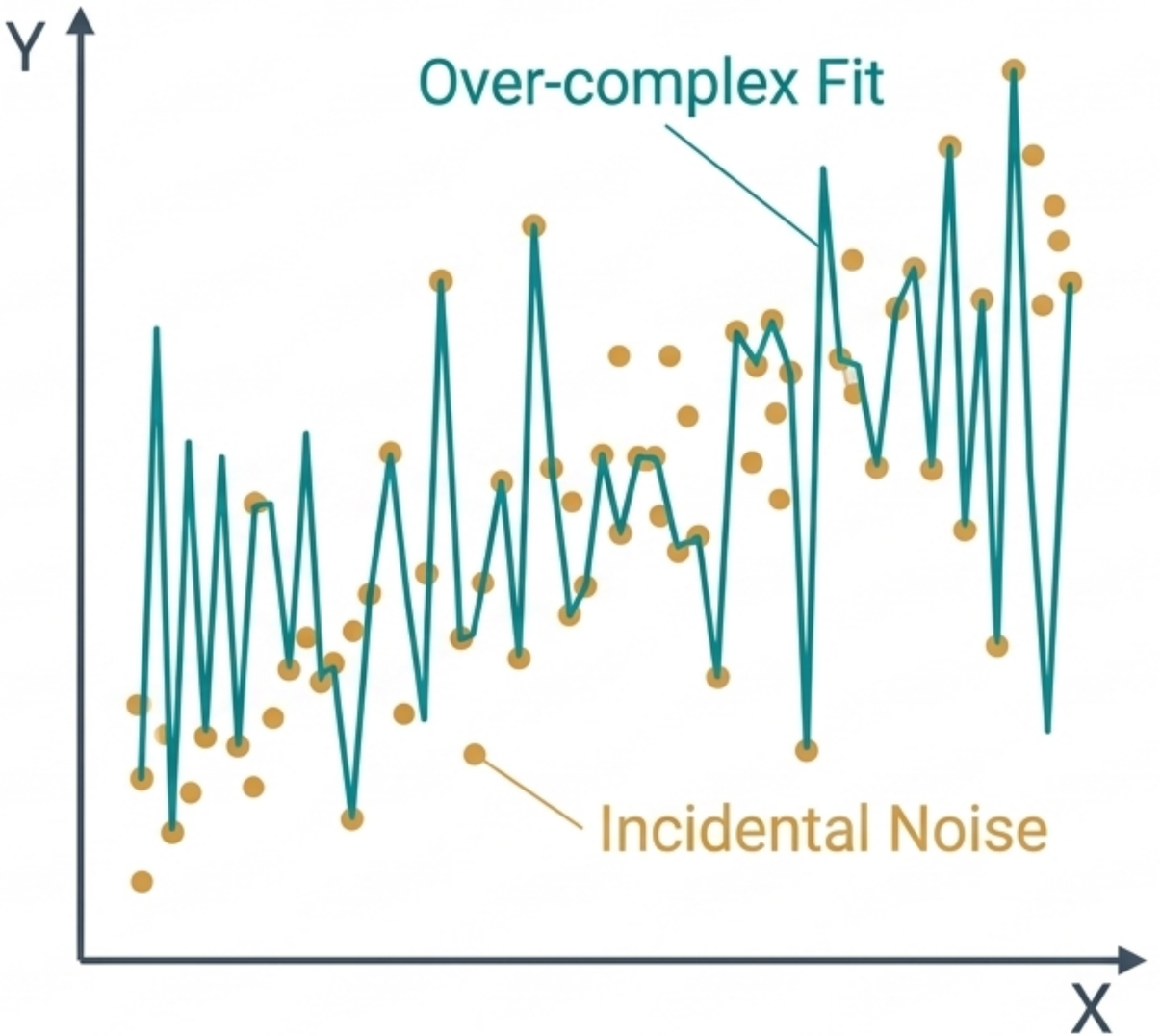
Overfitting Captures Noise and Fails to Generalize

Model learns training-specific variation instead of general patterns.

Algorithm perfectly captures incidental noise as significant features.

Training error is significantly lower than validation error.

Complex architectures, like deep trees, increase overfitting risk.



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Comparative Diagnostics for **Model Fitting**

Condition	Learned Pattern	Error Symptoms
Underfitting	Misses primary relationship	High training error, high validation error
Overfitting	Captures incidental noise	Low training error, high validation error

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2. Alternative Error Causes

Differentiate structural model failures from data-driven performance limits.

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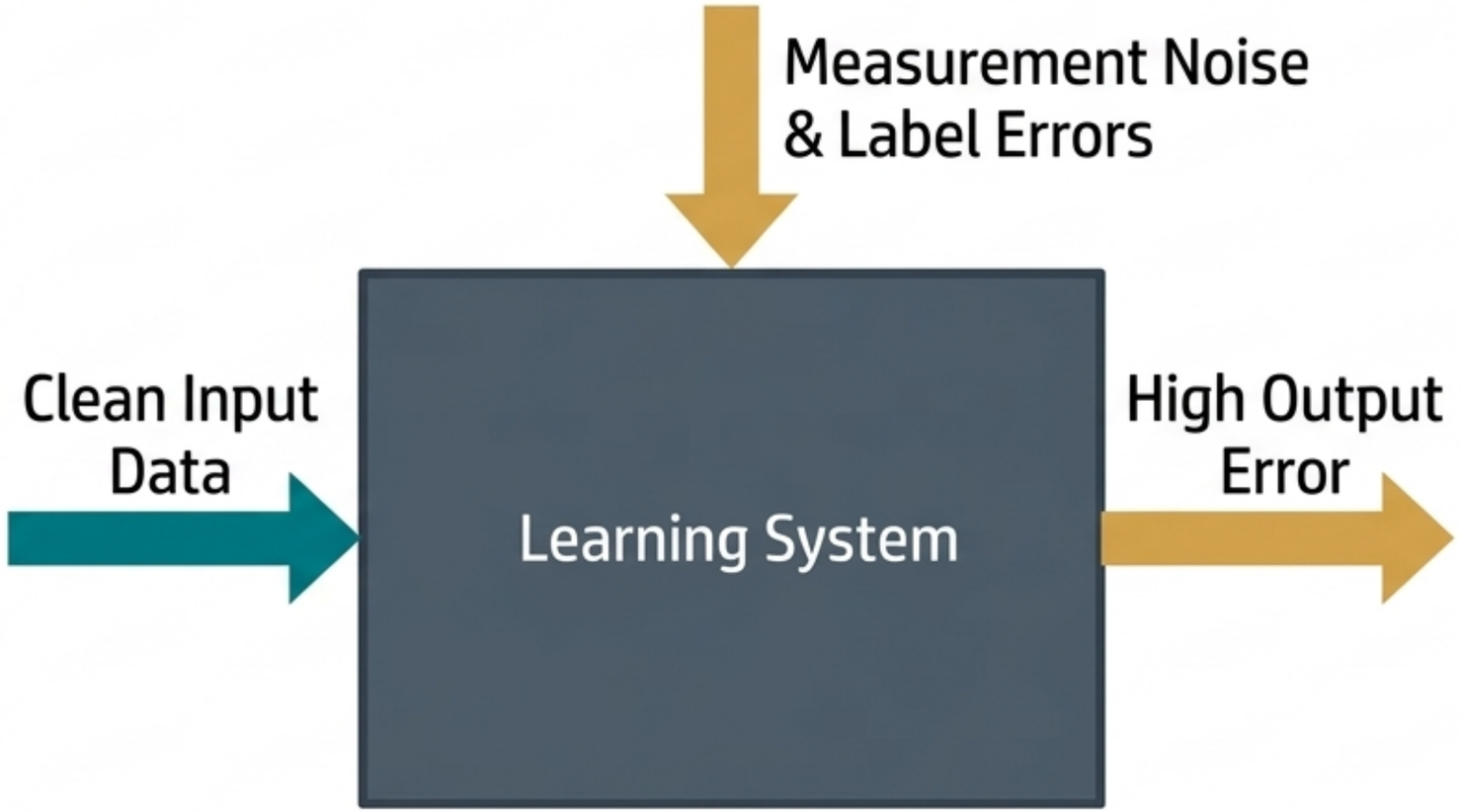
External Factors Inducing High System Error

High error does not strictly prove low model complexity.

Measurement noise intrinsically limits maximum achievable accuracy.

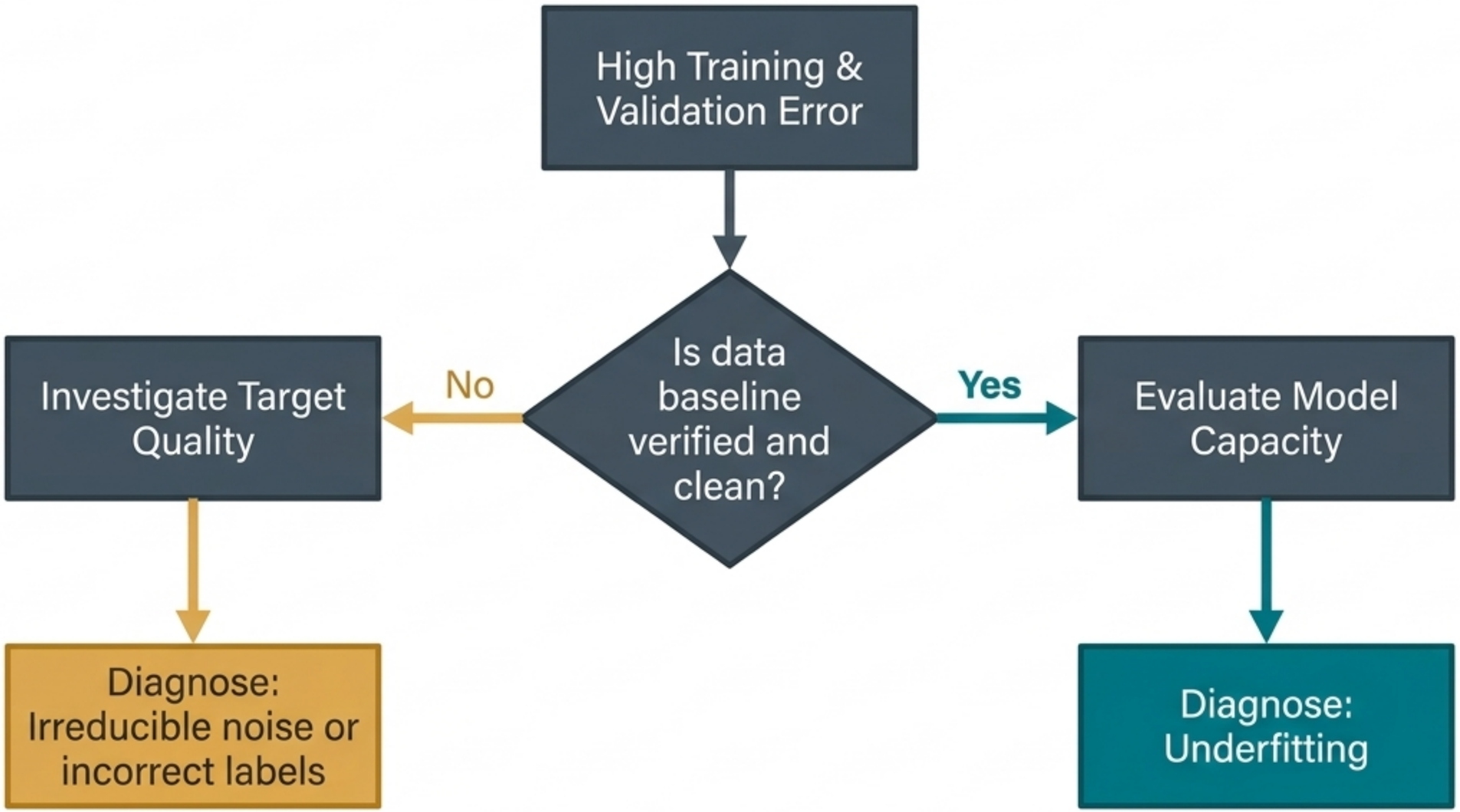
Incorrect target labels generate false error signals.

Missing input features prevent accurate relationship mapping.



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Decision Logic for High Dual-Set Error



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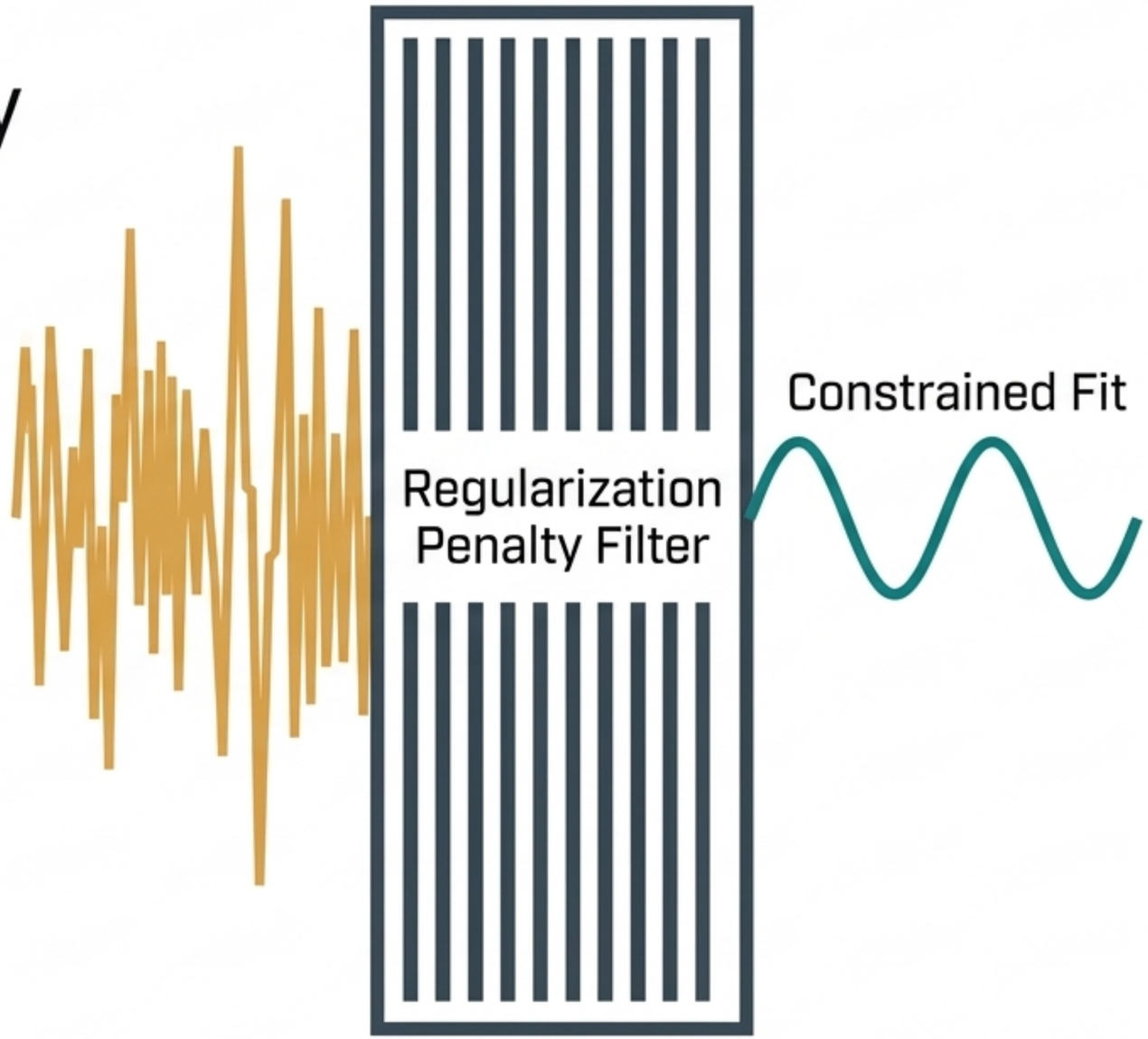
3. Remediation Strategies

Apply engineered controls to manage algorithmic complexity and training duration.

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Regularization Penalizes Excessive Model Complexity

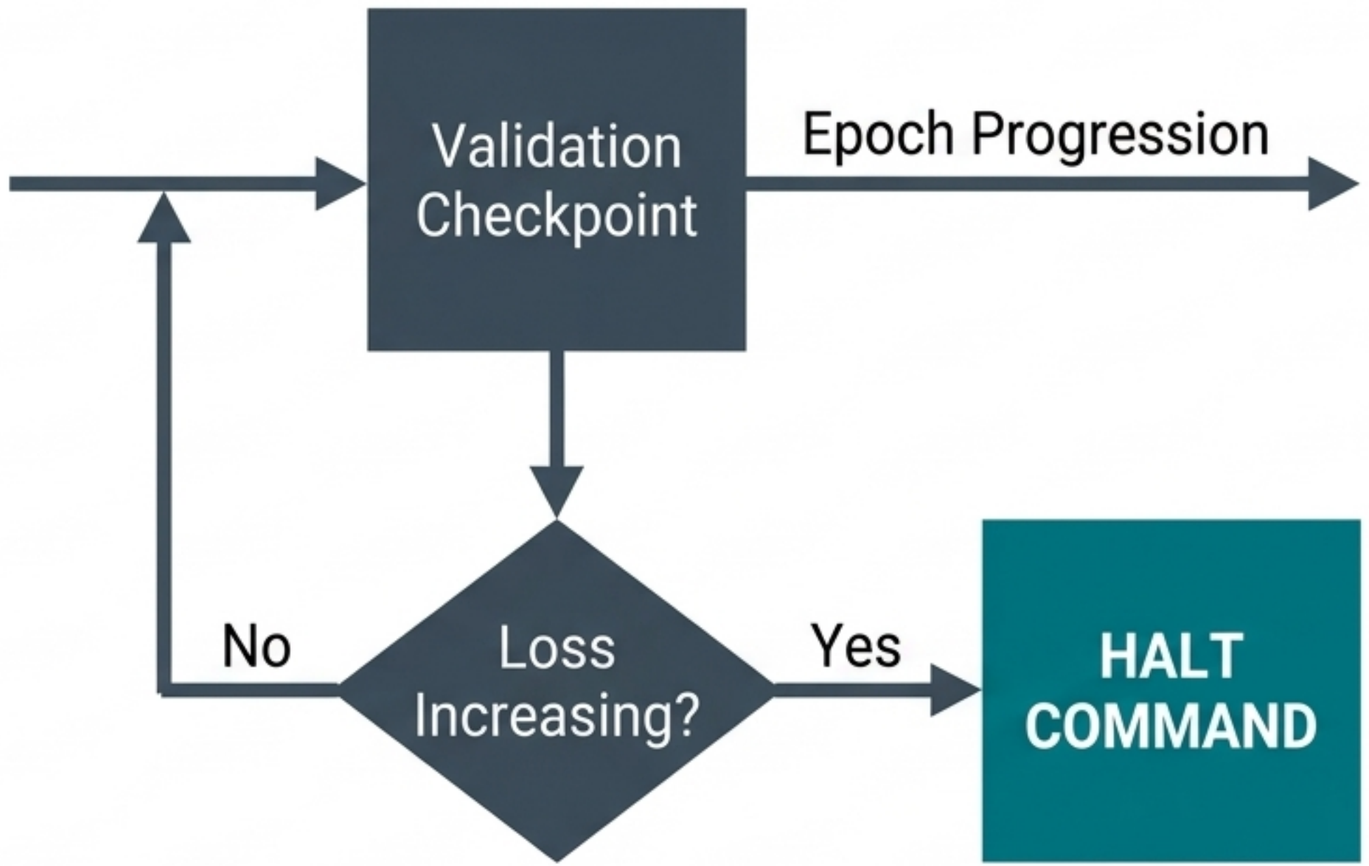
- Discourages unnecessarily complex algorithmic fitting patterns.
- Applies mathematical penalties to large fitted coefficients.
- Imposes strict structural constraints on model architecture.
- Reduces variance to improve generalized prediction accuracy.



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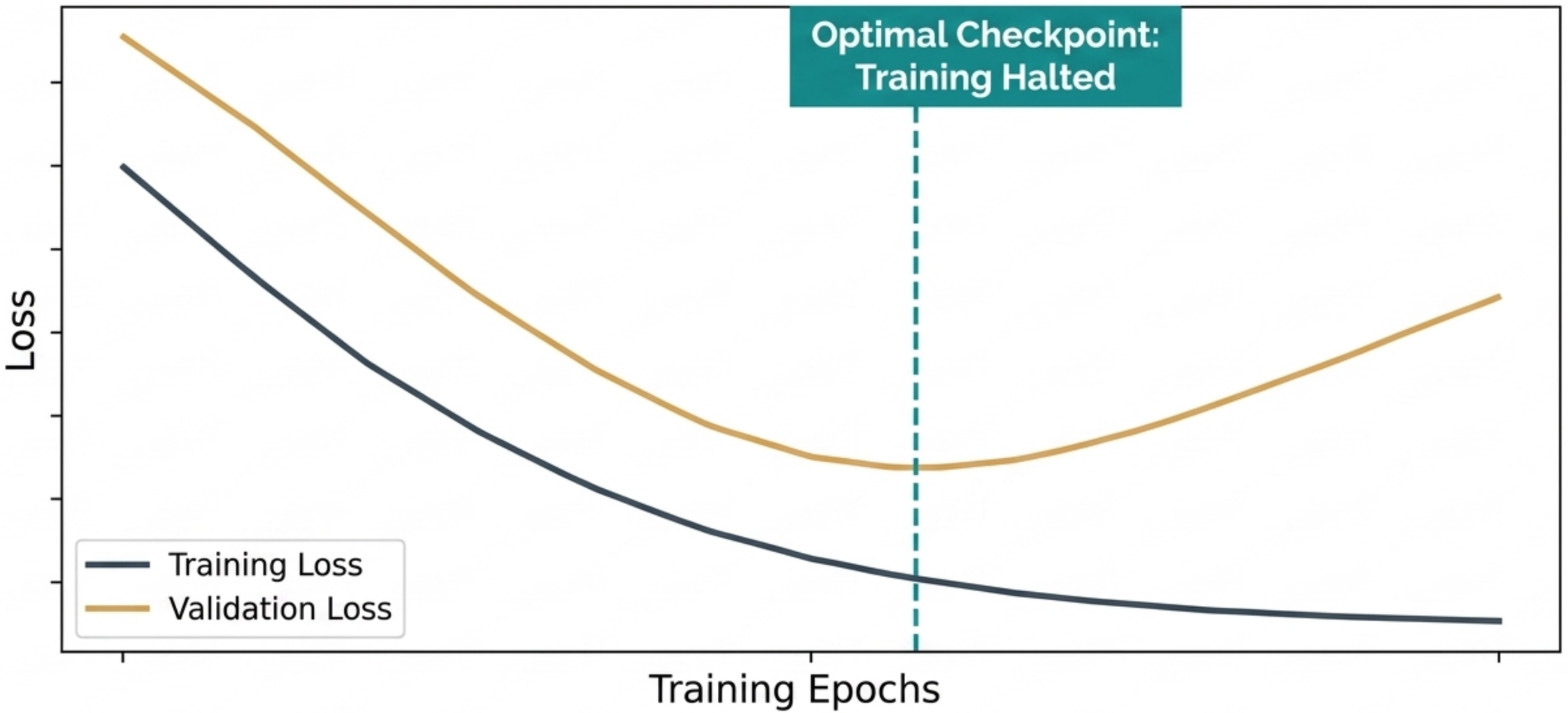
Early Stopping Halts Training Using Validation Data

- Selects an optimal training checkpoint using validation performance.
- Halts epoch progression before validation loss worsens.
- Operates exclusively within development data partitions.
- Never uses final-test scores to select training epochs.



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Validation Loss Trajectory Dictates Stop Point



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Key Points



1. Underfitting and overfitting represent failures to capture generalizable relationships.

2. High error rates require eliminating irreducible noise and label inaccuracy.

3. Regularization and early stopping engineer controls against excessive model complexity.

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